4.9 A point charge q is located in free space a distance d from the center of a dielectric sphere of radius a ($a < d$) and dielectric constant ϵ/ϵ_0.

a) Find the potential at all points in space as an expansion in spherical harmonics.

By symmetry, we may place the point charge on the z-axis at $z = d$. In this case, the problem is azimuthally symmetric, and we may expand the potential in Legendre polynomials instead of spherical harmonics. For the potential inside the dielectric sphere, we take

$$\Phi_{\text{in}} = \frac{q}{4\pi\epsilon} \sum_l \alpha_l \left(\frac{r}{a}\right)^l P_l(\cos\theta) \tag{9}$$

where the $q/4\pi\epsilon$ prefactor is taken for convenience (but can be absorbed into a redefinition of α_l if so desired). Note that we do not need any source term, since there are no charges inside the sphere. On the other hand, the solution outside the sphere is given by

$$\Phi_{\text{out}} = \frac{1}{4\pi\epsilon_0} \frac{q}{|\vec{x} - d\hat{z}|} + \Phi_0$$

where Φ_0 is a homogeneous solution to Laplace's equation, $\nabla^2\Phi_0 = 0$. Expanding in Legendre polynomials allows us to write

$$\Phi_{\text{out}} = \frac{q}{4\pi\epsilon_0} \sum_l \left[\frac{r_<^l}{r_>^{l+1}} + \beta_l \left(\frac{a}{r}\right)^{l+1}\right] P_l(\cos\theta) \tag{10}$$

Note that $r_< = \min(r, d)$ and $r_> = \max(r, d)$. Since we must match the parallel electric field and perpendicular electric displacement at $r = a$, we may take $r_< = r$ and $r_> = d$ when using Φ_{out} in the matching equations. For the parallel electric field, we have

$$E_\theta^{\text{in}} = -\frac{1}{r} \frac{\partial\Phi_{\text{in}}}{\partial\theta}\bigg|_{r=a} = \frac{q}{4\pi\epsilon} \sum_l \frac{\alpha_l}{a} P_l'(\cos\theta)\sin\theta$$

$$E_\theta^{\text{out}} = -\frac{1}{r} \frac{\partial\Phi_{\text{out}}}{\partial\theta}\bigg|_{r=a} = \frac{q}{4\pi\epsilon_0} \sum_l \left[\frac{a^{l-1}}{d^{l+1}} + \frac{\beta_l}{a}\right] P_l'(\cos\theta)\sin\theta$$

Matching these gives

$$\alpha_l = \frac{\epsilon}{\epsilon_0}\left[\frac{a^l}{d^{l+1}} + \beta_l\right] \tag{11}$$

On the other hand, for the perpendicular electric displacement, we have

$$D_r^{\text{in}} = -\epsilon \left.\frac{\partial \Phi_{\text{in}}}{\partial r}\right|_{r=a} = \frac{q}{4\pi}\sum_l \frac{l\alpha_l}{a} P_l(\cos\theta)$$

$$D_r^{\text{out}} = -\epsilon_0 \left.\frac{\partial \Phi_{\text{out}}}{\partial r}\right|_{r=a} = \frac{q}{4\pi}\sum_l \left[\frac{la^{l-1}}{d^{l+1}} - \frac{(l+1)\beta_l}{a}\right] P_l(\cos\theta)$$

Matching gives

$$\alpha_l = \frac{a^l}{d^{l+1}} - \frac{(l+1)}{l}\beta_l \tag{12}$$

Solving (11) and (12) yields

$$\alpha_l = \frac{2l+1}{l + \frac{\epsilon_0}{\epsilon}(l+1)}\frac{a^l}{d^{l+1}}$$

$$\beta_l = \frac{(\frac{\epsilon_0}{\epsilon} - 1)l}{l + \frac{\epsilon_0}{\epsilon}(l+1)}\frac{a^l}{d^{l+1}}$$

As a result, the interior and exterior potential, given by (9) and (10), has the form

$$\begin{aligned} \Phi_{\text{in}} &= \frac{q}{4\pi\epsilon d}\sum_l \frac{2l+1}{l + \frac{\epsilon_0}{\epsilon}(l+1)}\left(\frac{r}{d}\right)^l P_l(\cos\theta) \\ \Phi_{\text{out}} &= \frac{q}{4\pi\epsilon_0}\sum_l \left[\frac{r_<^l}{r_>^{l+1}} + \frac{(\frac{\epsilon_0}{\epsilon} - 1)l}{l + \frac{\epsilon_0}{\epsilon}(l+1)}\frac{a^{2l+1}}{(rd)^{l+1}}\right] P_l(\cos\theta) \end{aligned} \tag{13}$$

Note that, without the dielectric sphere (so that $\epsilon = \epsilon_0$), this reduces to

$$\Phi_{\text{in}} = \frac{q}{4\pi\epsilon_0}\sum_l \frac{r^l}{d^{l+1}} P_l(\cos\theta)$$

$$\Phi_{\text{out}} = \frac{q}{4\pi\epsilon_0}\sum_l \frac{r_<^l}{r_>^{l+1}} P_l(\cos\theta)$$

which is simply the free space result

$$\Phi = \frac{q}{4\pi\epsilon_0|\vec{x} - d\hat{z}|}$$

b) Calculate the rectangular components of the electric field *near* the center of the sphere.

Near the center, we may expand Φ_{in} in (13).

$$\begin{aligned}
\Phi_{\text{in}} &= \frac{q}{4\pi\epsilon d}\left[\frac{1}{\frac{\epsilon_0}{\epsilon}} + \frac{3}{1+2\frac{\epsilon_0}{\epsilon}}\frac{r}{d}\cos\theta + \frac{5}{2+3\frac{\epsilon_0}{\epsilon}}\left(\frac{r}{d}\right)^2\frac{3\cos^2\theta-1}{2} + \cdots\right] \\
&= \frac{q}{4\pi\epsilon_0 d}\left[1 + \frac{3}{2+\frac{\epsilon}{\epsilon_0}}\frac{z}{d} + \frac{5}{3+2\frac{\epsilon}{\epsilon_0}}\frac{3z^2-r^2}{2d^2} + \cdots\right]
\end{aligned}$$

The electric field is then

$$\vec{E} = -\vec{\nabla}\Phi_{\text{in}} = \frac{q}{4\pi\epsilon_0 d^2}\left[-\frac{3}{2+\frac{\epsilon}{\epsilon_0}}\hat{z} + \frac{5}{3+2\frac{\epsilon}{\epsilon_0}}\frac{x\hat{x}+y\hat{y}-2z\hat{z}}{d} + \cdots\right]$$

Very close to the center, the field is nearly uniform, and pointed in the $-\hat{z}$ direction (assuming $q > 0$). The presence of the dielectric modifies the point charge result $\vec{E} = -(q/4\pi\epsilon_0 d^2)\hat{z}$ by the factor $3/(2+\epsilon/\epsilon_0)$.

c) Verify that, in the limit $\epsilon/\epsilon_0 \to \infty$, your result is the same as that for the conducting sphere.

For $\epsilon/\epsilon_0 \to \infty$, the potential (13) reduces to

$$\begin{aligned}
\Phi_{\text{in}} &= \frac{q}{4\pi\epsilon_0 d} \\
\Phi_{\text{out}} &= \frac{q}{4\pi\epsilon_0}\left[\sum_{l=0}^{\infty}\frac{r_<^l}{r_>^{l+1}} - \sum_{l=1}^{\infty}\frac{a^{2l+1}}{(rd)^{l+1}}\right]P_l(\cos\theta) \\
&= \frac{q(a/d)}{4\pi\epsilon_0 r} + \frac{q}{4\pi\epsilon_0}\sum_{l=0}^{\infty}\left(r_<^l - \frac{a^{2l+1}}{r_<^{l+1}}\right)\frac{1}{r_>^{l+1}}P_l(\cos\theta) \\
&= \frac{q}{4\pi\epsilon_0}\left[\frac{a/d}{r} + \frac{1}{|\vec{x}-d\hat{z}|} - \frac{a/d}{|\vec{x}-(a^2/d)\hat{z}|}\right]
\end{aligned}$$

which is indeed the correct result for a conducting sphere. Note that the $l = 0$ term in the sum had to be treated with care when taking the limit $\epsilon/\epsilon_0 \to \infty$. This results in the $q(a/d)/4\pi\epsilon_0 r$ term corresponding to an uncharged (and ungrounded) conducting sphere having non-zero potential when the charge q is brought near it.